Project Documentation

Planning & Management · Literature Review · Requirements Gathering

Digital Egypt Pioneers Initiative (DEPI) | Submission: 16 Oct 2026

1. Project Planning & Management

PROJECT PROPOSAL

Delivered.

PROJECT PLAN

Phase	Focus
1	Defining problem scenarios, data understanding, and establishing the core architecture contract.
2	Building the web skeleton, developing sign-recognition and CV models, and running language/speech trials.
3	Connecting the inference interface, profile retrieval system, and adaptive routing logic.
4	Functional testing, assessing communication quality, and refining text outputs.
5	Delivering the integrated website, running end-to-end demonstrations, and documenting system limits.

TASK ASSIGNMENT & ROLES

Member	Responsibility	Tools
1	Sign Language & Gesture CV Track	MediaPipe / OpenPose
2	Object Detection & OCR CV Track	YOLOv8 / EasyOCR
3	NLP & Text Normalization	LLM semantic cleanup
4	Data Engineering & Academic Dataset Sourcing	—
5	Adaptive Routing & Architecture Engine	—
6	Model Testing & Semantic Evaluation	—

RISK ASSESSMENT & MITIGATION

Risk	Mitigation
High real-time latency	Lightweight Edge AI and client-side processing to prevent WebSocket bottlenecks.
LLM hallucinations / intent loss	Open-weight models (Qwen) restricted strictly to semantic cleanup: “improve form, preserve information.”
Scope creep	Target a prototype web ecosystem rather than unconstrained production software.
Data scarcity	Source open-source datasets via academic databases (IEEE, Springer).

KPIS

Metric	What it measures
Inference response time	Speed of CV and LLM processing.
Routing accuracy	Correct selection of transformation pathways.
Semantic preservation rate	Names, numbers, dates, and actions kept without distortion.
System reliability	Stability of backend routing and extension uptime.
Accessibility accuracy	End-to-end success of receivers understanding translated messages.

2. Literature Review

FEEDBACK & EVALUATION

- **Instructor insights:** initial review notes stress the importance of targeting multi-modal communication gaps.
- **Key recommendations:** panel suggestions integrated, such as adding currency recognition to support blind users.

SUGGESTED IMPROVEMENTS

- **Technical:** lightweight open-weight models (Qwen) for robust semantic text normalization.
- **Usability & architecture:** optimize Edge AI and WebSocket streaming to minimize latency during live interactions.

FINAL GRADING CRITERIA

Criterion	Assessed on
Documentation	Project proposals, literature reviews, and requirements gathering.
Technical implementation	Computer vision tracks, NLP pipelines, and the adaptive routing engine.
Testing reports	Model accuracy, semantic preservation, and performance metrics.
Final presentation	Live end-to-end multi-modal demonstration and clarity of explanation.

3. Requirements Gathering

STAKEHOLDER ANALYSIS

Stakeholder	Needs
Deaf / mute users	Real-time sign language translation and live captions during communication.
Blind users	Object/hazard detection, spatial audio, and currency recognition.

USER STORIES & USE CASES

- **Deaf user:** “I want incoming voice call audio automatically transcribed into live text captions.”
- **Mute user:** “I want my sign language gestures captured by the camera and translated into speech/text instantly.”
- **Blind user:** “I want objects and banknotes described via spatial audio and object detection.”

FUNCTIONAL REQUIREMENTS

Component	Function
MediaPipe / OpenPose	Tracks hand, arm, and facial landmarks for continuous sign language translation and lip-reading.
YOLOv8	Object detection, depth estimation, spatial audio cues, and currency recognition for banknotes and coins.
Qwen LLM	Semantic text normalization (“improve form, preserve information”).
RAG (optional)	Optional enhancement: retrieval of verified accessibility guidelines, added if time allows.

NON-FUNCTIONAL REQUIREMENTS

Area	Requirement
Performance	Low-latency streaming via WebSockets for real-time communication.
Security & privacy	Localized processing via Edge AI to protect user data and privacy.
Reliability	Stable backend routing and robust extension uptime during multi-modal sessions.
UI/UX usability	Dynamic interface modifications matching the selected user accessibility profile.